(i) Constant function

A function $f : A \to B$ is called a constant function if the range of f contains only one element. That is, $f(x) = c$, for all $x \in A$ and for some fixed $c \in B$.

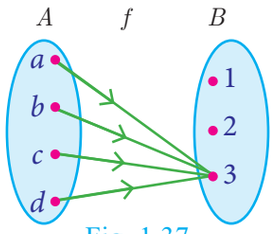


Fig. 1.37

Illustration 16

From Fig.1.37, $A = \{a, b, c, d\}$, $B = \{1, 2, 3\}$ and $f = \{(a, 3), (b, 3), (c, 3), (d, 3)\}$. $\because f(x) = 3 \ \forall \ x \in A$, Range of $f = \{3\}$, f is a constant function.

(ii) Identity function

Let A be a non–empty set. Then the function $f : A \to A$ defined by $f(x) = x$ for all $x \in A$ is called an identity function on A and is denoted by I_A.

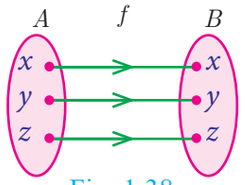


Fig. 1.38

Illustration 17

If $A = \{a, b, c\}$ then $f = I_A = \{(a, a), (b, b), (c, c)\}$ is an identity function on A.

Thinking Corner

Is an identity function one to one function?

(iii) Real valued function

A function $f : A \to B$ is called a real valued function if the range of f is a subset of the set of all real numbers $\mathbb{R}$. That is, $f(a) \subseteq \mathbb{R}, \ \forall \ a \in A$.

Progress Check

State True or False.

1. All one – one functions are onto functions.
2. There will be no one – one function from A to B when $n(A) = 4$, $n(B) = 3$.
3. All onto functions are one – one functions.
4. There will be no onto function from A to B when $n(A) = 4$, $n(B) = 5$.
5. If f is a bijection from A to B, then $n(A) = n(B)$.
6. If $n(A) = n(B)$, then f is a bijection from A to B.
7. All constant functions are bijections.

Example 1.17 Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = 3x - 5$. Find the values of a and b given that $(a, 4)$ and $(1, b)$ belong to f.

Solution $f(x) = 3x - 5$ can be written as $f = \{(x, \ 3x - 5) \mid x \in R\}$

$(a, 4)$ means the image of a is 4. i.e., $f(a) = 4$

$3a - 5 = 4 \Rightarrow a = 3$

$(1, b)$ means the image of 1 is b. i.e., $f(1) = b$

$3(1) - 5 = b \Rightarrow b = -2$

1 Relation and Functions.indd 23 15-12-2020 02:50:22 PM